Dose-response (input-output) curves:

Ground truth for testing TMS methods?

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citing work with / by

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How do you..?

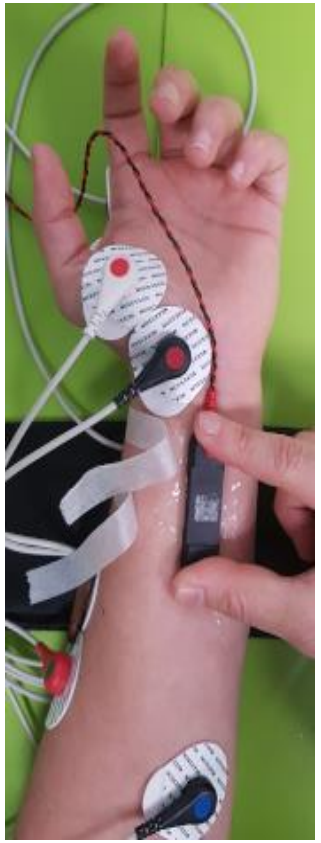
1. Know if your TMS methods are working?

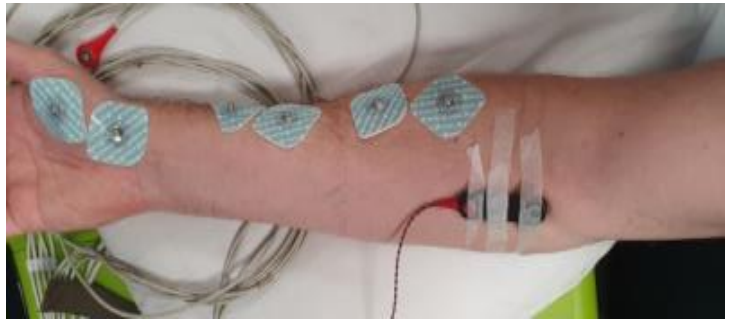
2. Test whether a manipulation has an effect?

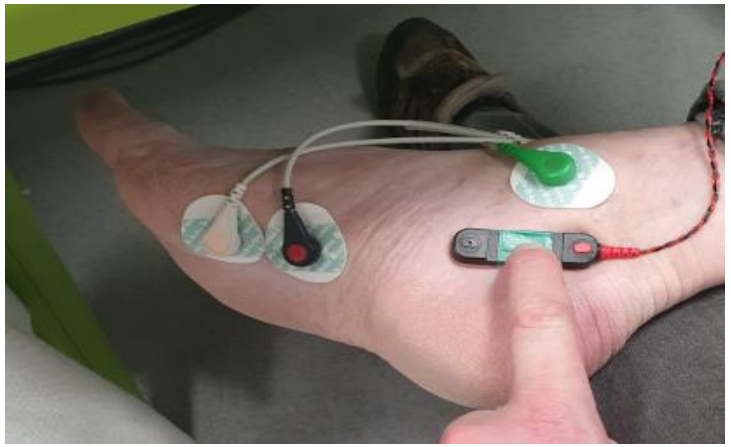
3. Calibrate methods & results across labs?

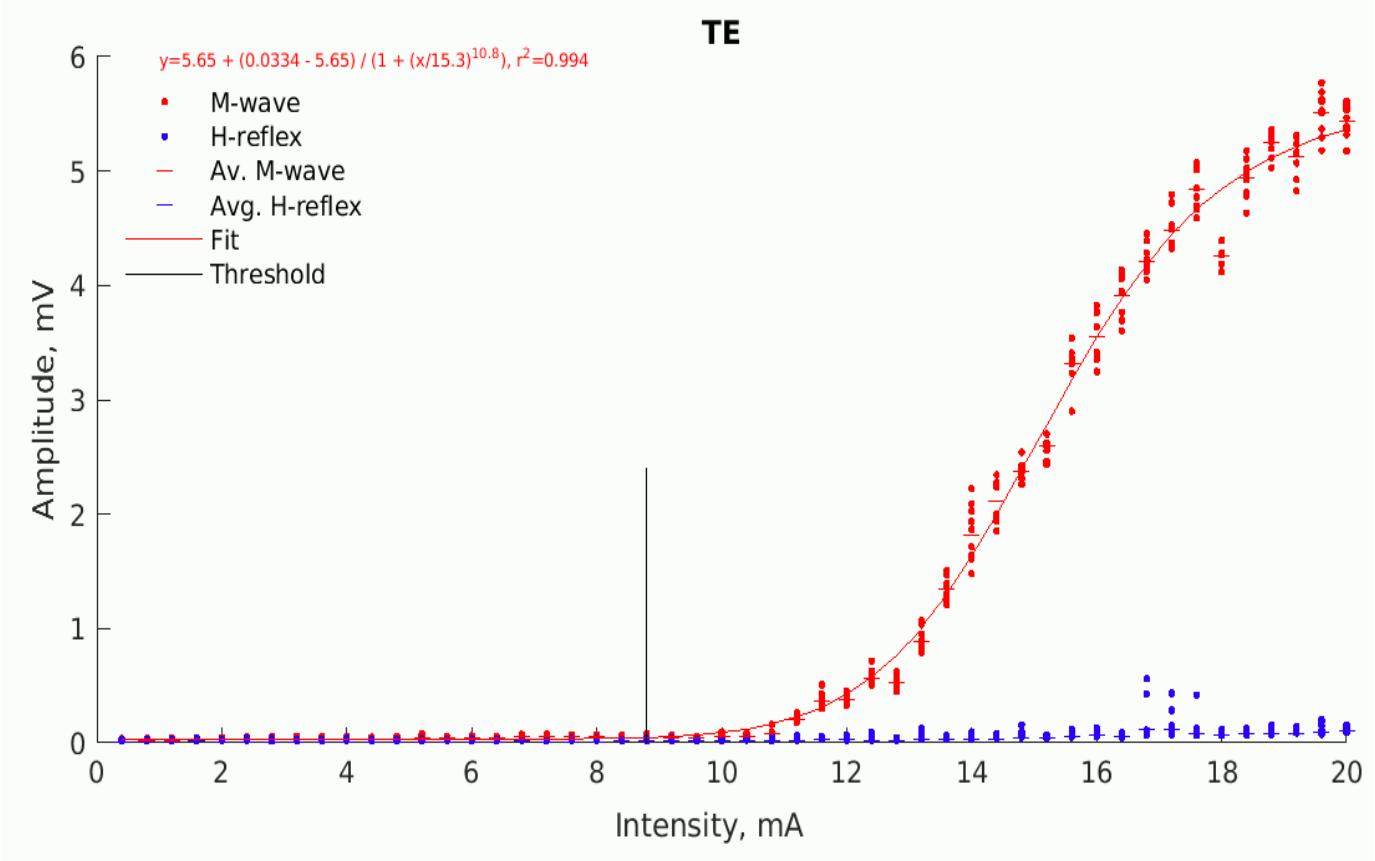
The (peripheral) motor system provides ground truth

- Stimulate the peripheral motor nerve at increasing intensities
- Very lawful increase in motor-unit recruitment, up to MMAX



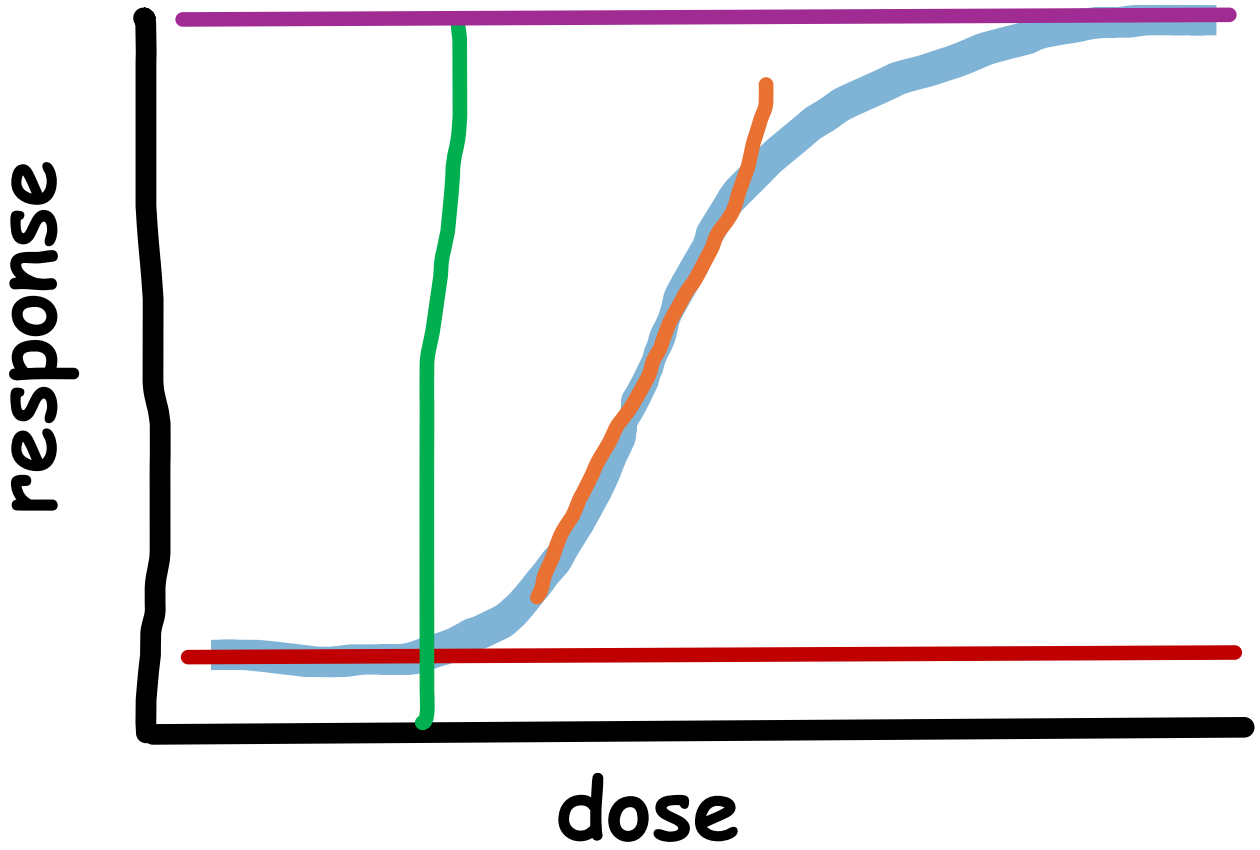






Model

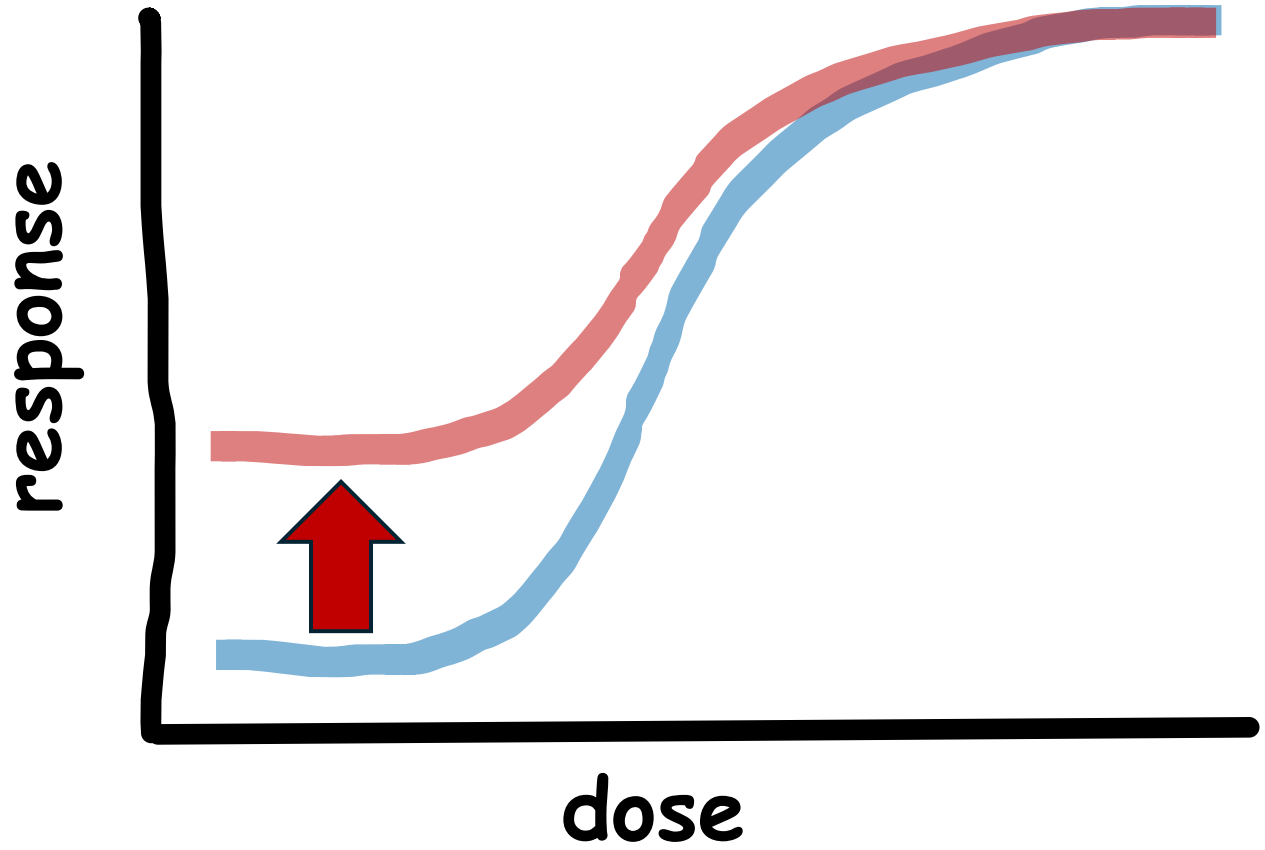
Neuromuscular output is a sigmoid function of stimulus intensity



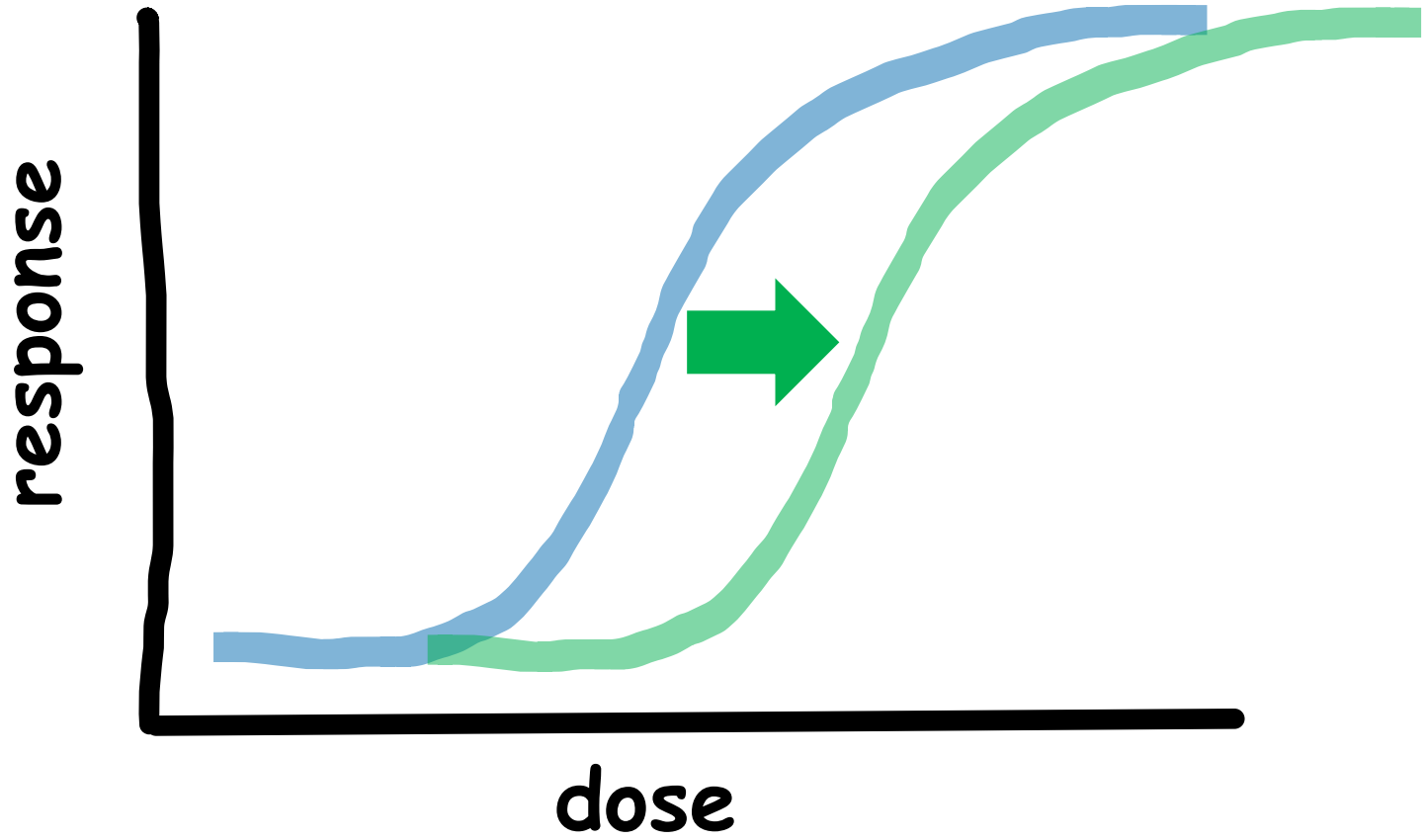
Parameters:

- Baseline (y)
- Threshold (x)
- Growth rate (y/x)
- Plateau (y)

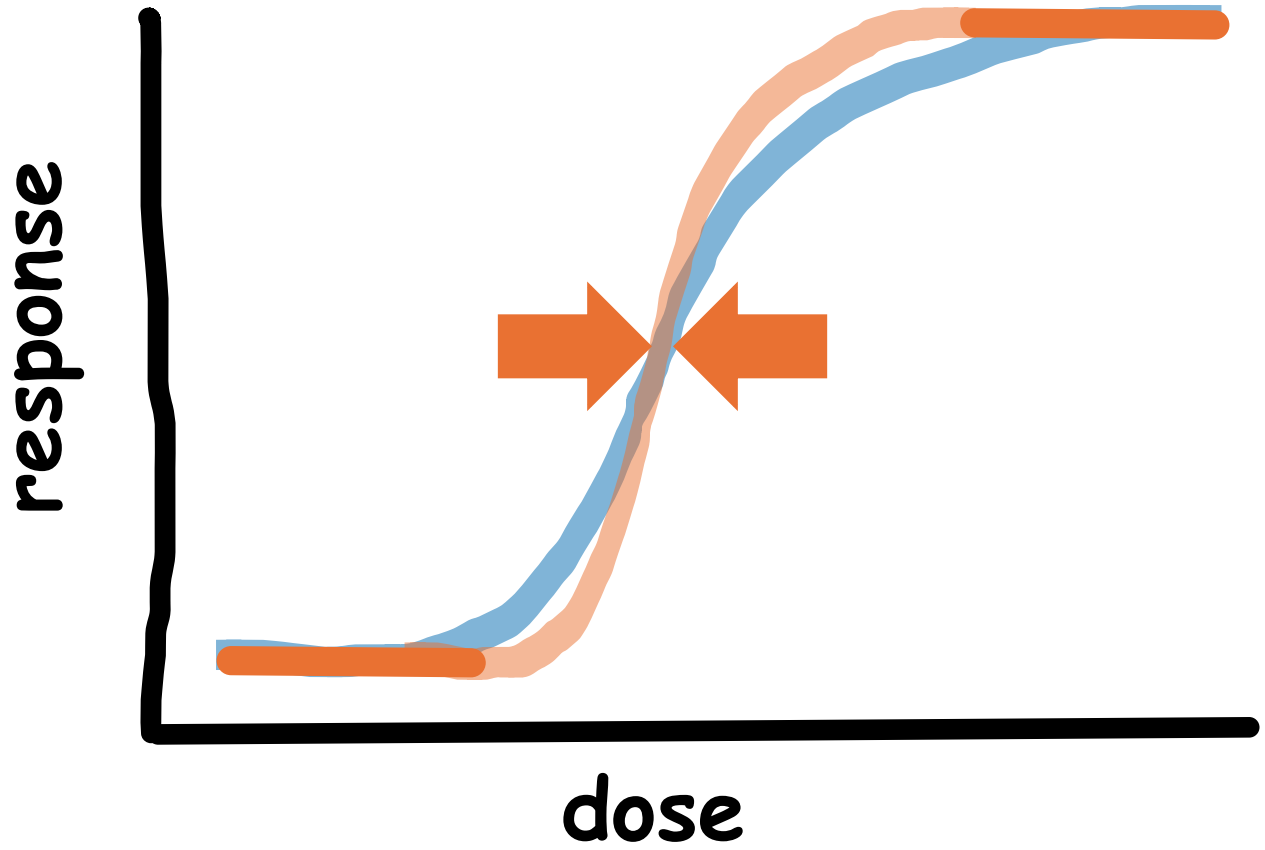
Theory predicts baseline (y) increase



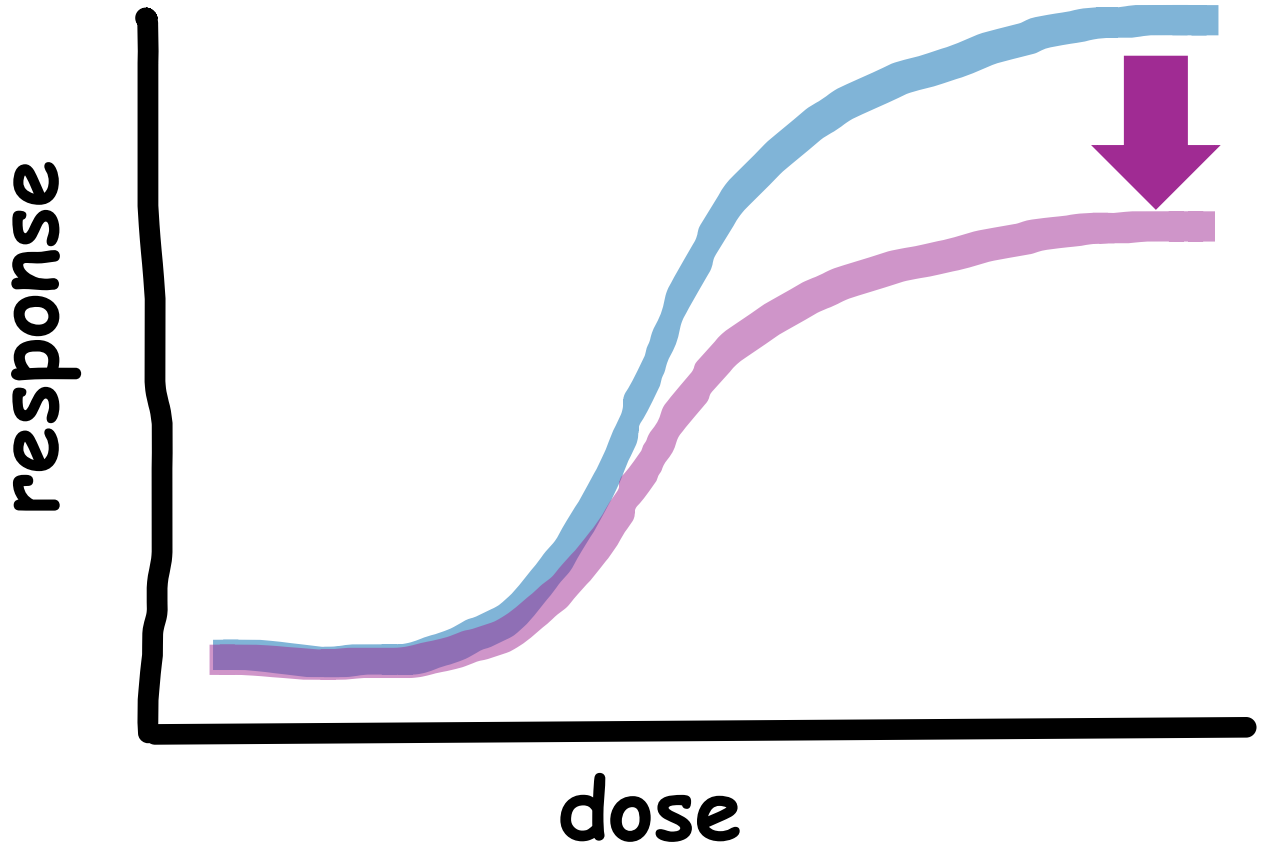
Theory predicts threshold (x) increase



Theory predicts growth (y/x) increase

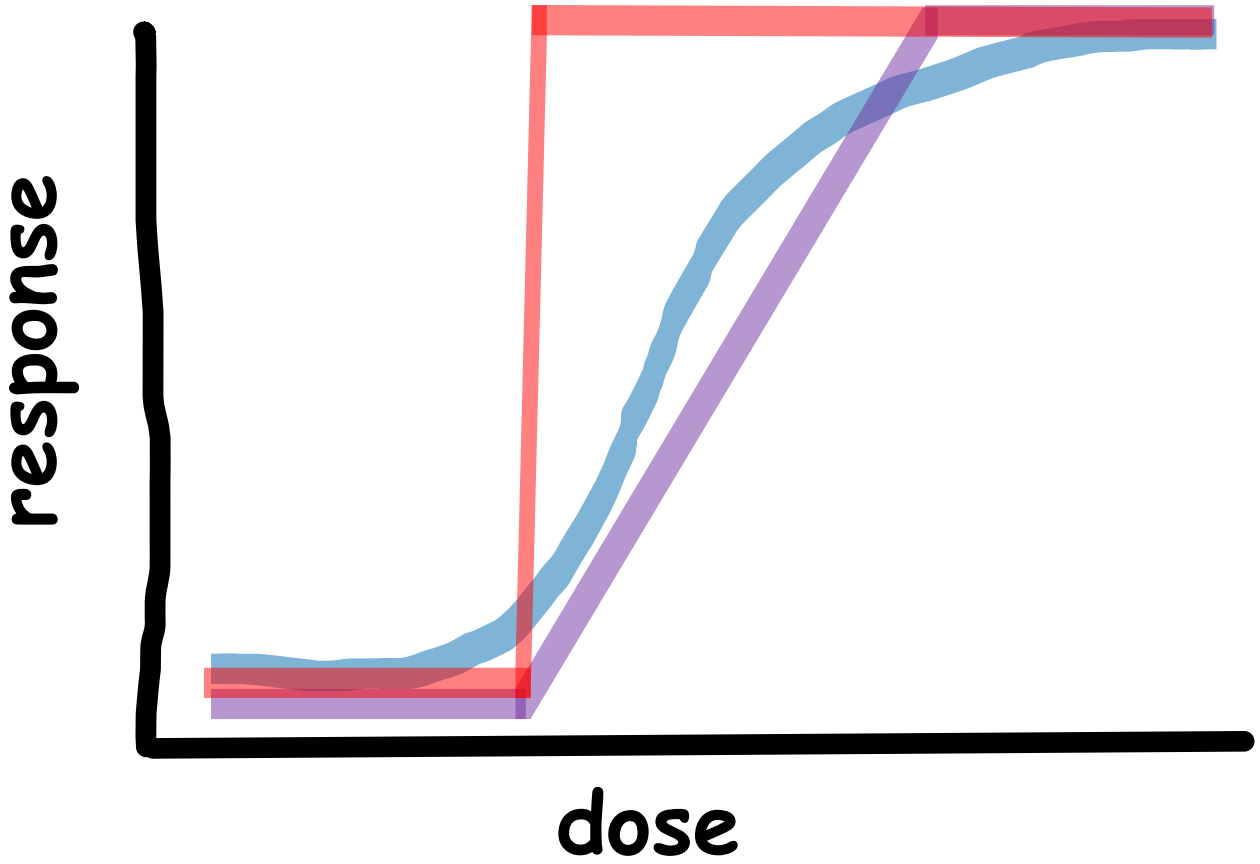


Theory predicts plateau (y) decrease



Testing theories with (motor) TMS

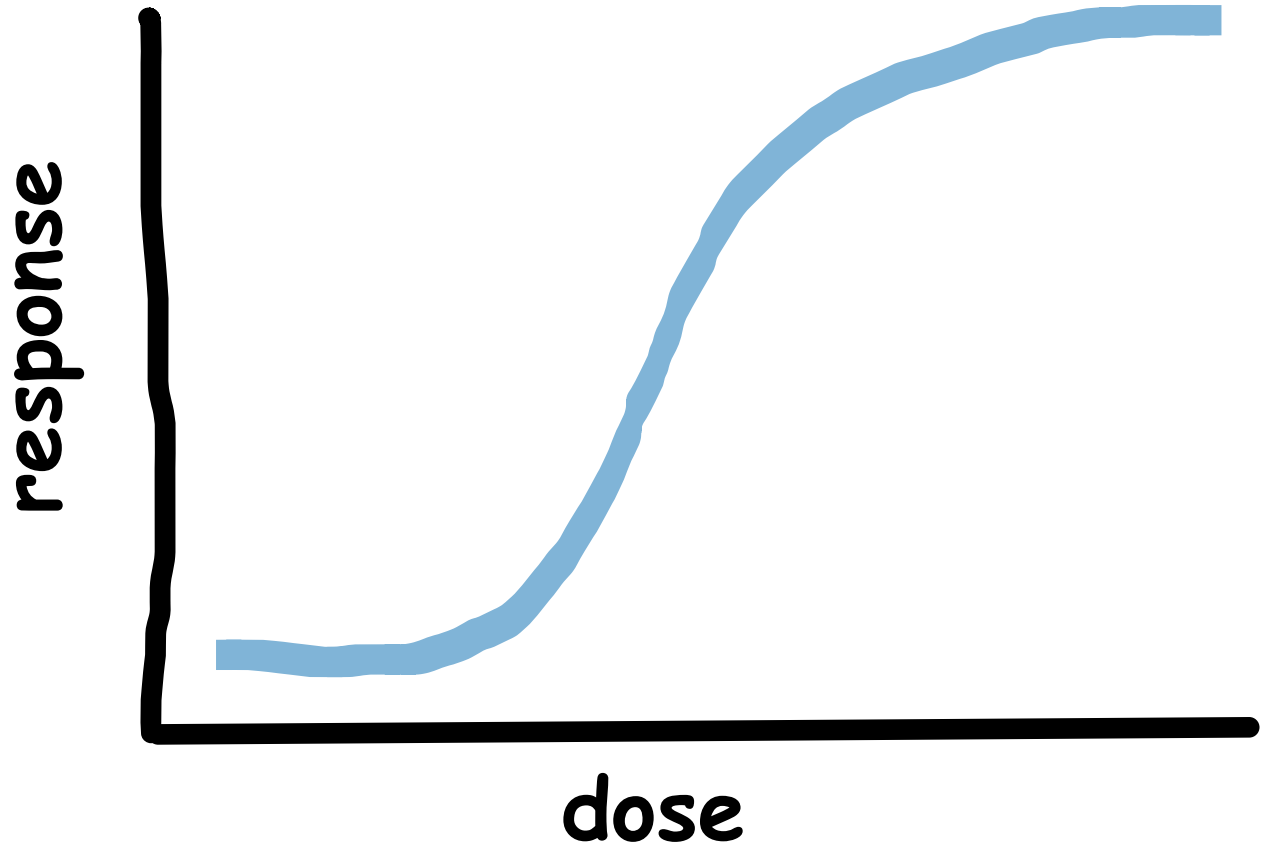
Corticospinal **output is a step, linear OR non-linear function of excitatory input**



Feature	Step	Linear	Non-linear
Threshold	Yes	Yes	Yes
Velocity	Instant	Constant	Bell
Acceleration	Instant	Instant	Wave
Plateau	Yes	Yes	Yes

Model

Working hypothesis: Corticospinal output is a sigmoid function of TMS intensity

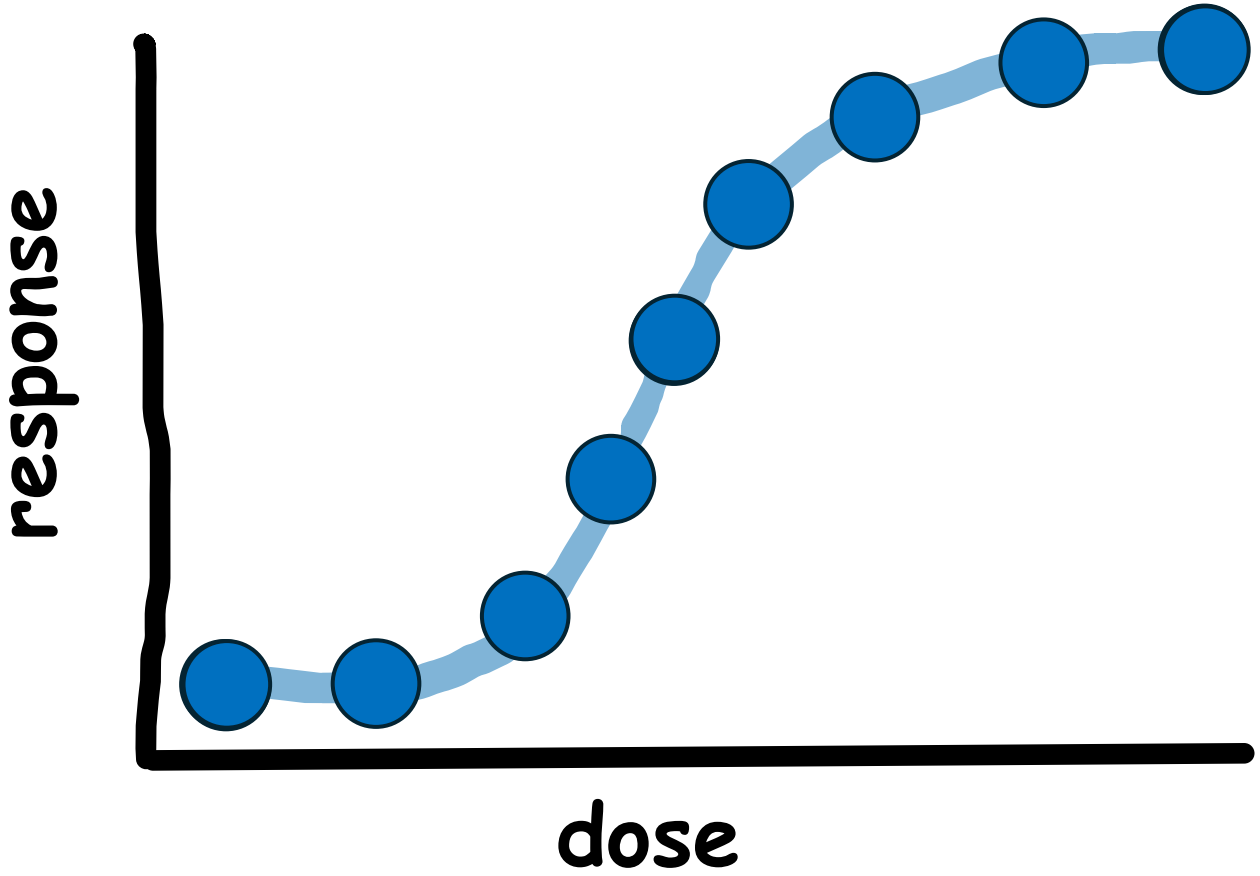


Predictions:

- Threshold ~50% MSO
- Increase ~1mV / 5% MSO
- Plateau ~80% MSO
- ~Sigmoid
- ~Continuous

Model testing

Working hypothesis: Corticospinal output is a sigmoid function of TMS intensity



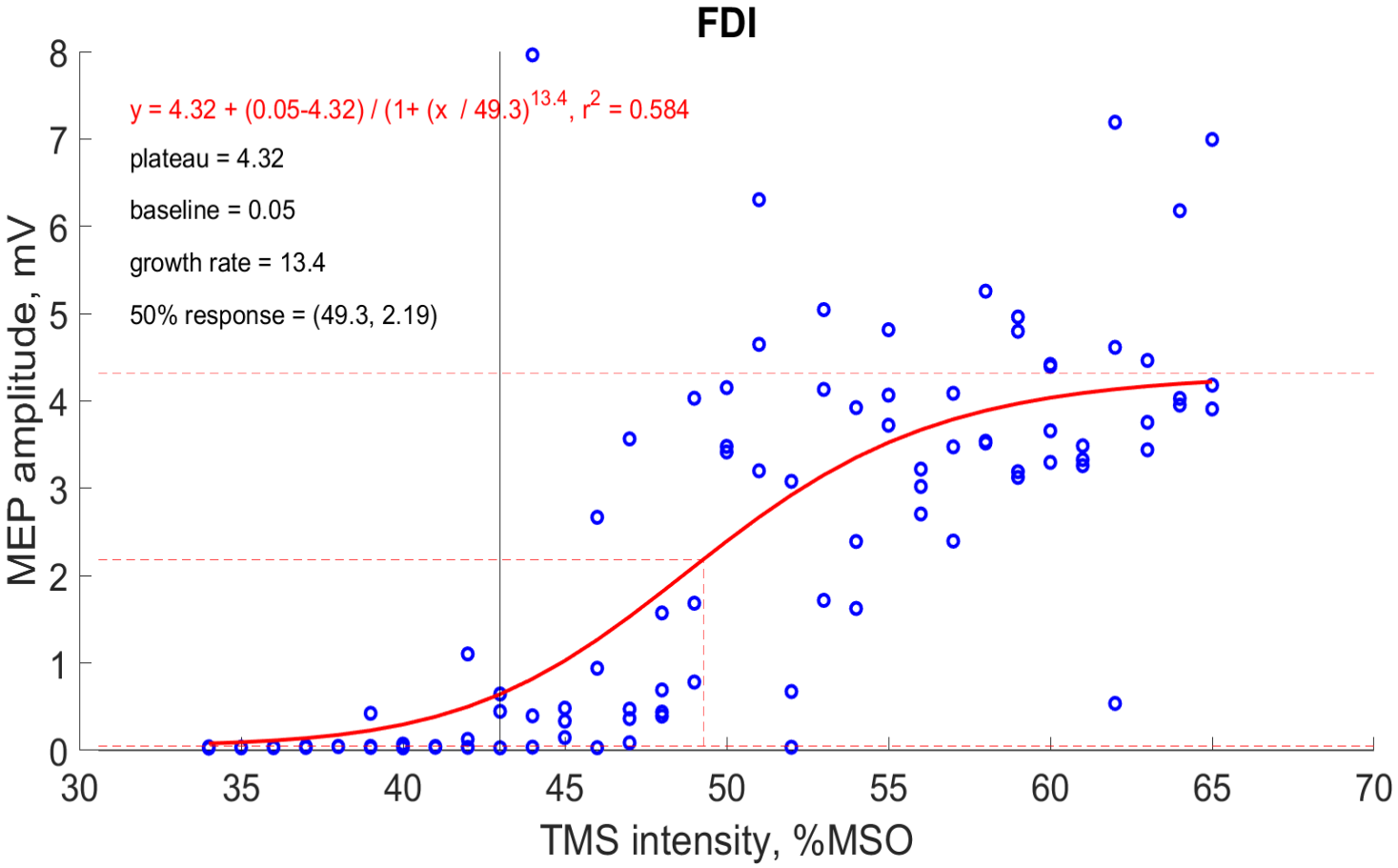
***Minimal:* 2** intensities
Can't discriminate models

Intermediate: 3 intensities
Can't discriminate linear/non

Full: ALL intensities
Can discriminate linear/non

Data

Working hypothesis: Corticospinal output is a sigmoid function of TMS intensity



Design:

- Discrete (6) vs. Continuous (30)

Baseline:

- Ignored vs. Corrected

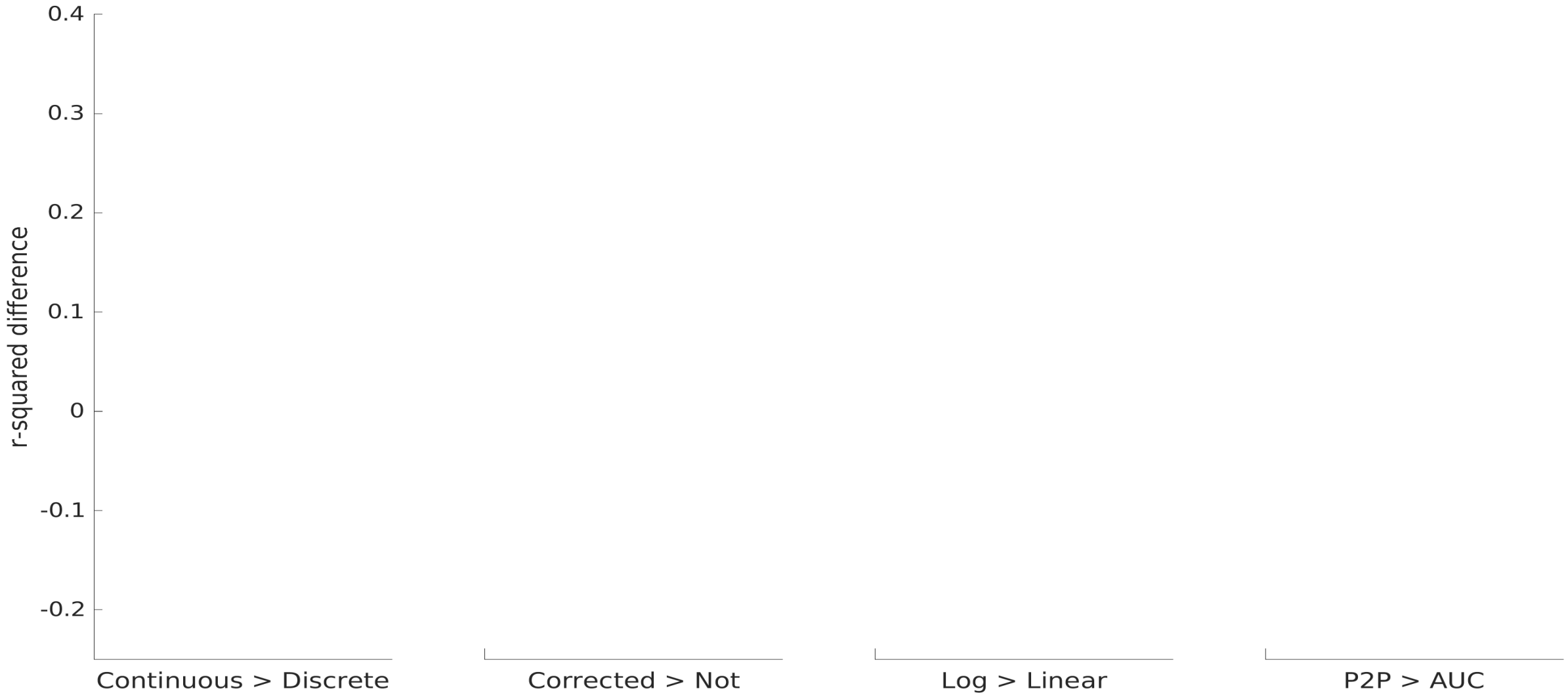
Transform

- None vs. logarithmic

Measure:

- P-2-P vs. A-U-C

Data



Best fit = **continuous design, baseline-corrected, log-transformed, peak-to-peak MEPs**

So..?

- **Test TMS methods using D-R curve fits (R^2) as a DV**
  - Background EMG activity
  - MEP measurements
  - Effects of motor preparation, conflict, imagery?
  - Cognitive effects on motor system
  - More widely? Then we need other dose-response curves

Discussion: Would this work for your study?

Dose-response code (MATLAB) is on the github